

Games Engineering II – CW208

Module Information (CW208)

- **Module Title:** Games Engineering II
- **Credits:** 10
- **Level:** 8 (Final year undergraduate)
- **Languages Used:** C++11/14, Erlang, Haskell

Learning Outcomes

By completing this module, students will be able to:

1. Select, design, and implement testing strategies appropriate to games development.
Evaluate methods for synchronising concurrent processes and their effects in game execution environments.
Design algorithms that execute on multiple cores or processors.
Develop functional programs and integrate them into digital games.
Apply suitable design patterns in games.

Assessment Breakdown

- Continuous Assessment: **20%**
[One Class Test (10 Marks) and Two Assignments (5 Marks each)]
- Project: **20%**
[Student selected individual project, weighing: Code (10 Marks), Report (10 Marks)]
- Practical labs: **20%**
[10 Labs weighing (2 Marks) each]
- Final written exam: **40%**

Please Note:

All late submissions will impose a 50% Marks reduction penalty.

You can send me a blank email with subject (**#mygrades**) to receive your grades automatically.

Recommended Books

1. *Modern C++ Programming with Test-Driven Development* – Jeff Langr
 2. *Game Programming Patterns* – Robert Nystrom
 3. *Game Engine Architecture* – Jason Gregory
 4. *C++ Concurrency in Action* – Anthony Williams
 5. *Parallel and Concurrent Programming in Haskell* – Simon Marlow
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GIT Student Setup

1. Clone the repository

```
git clone https://github.com/oa-SETU/GamesENGIIassignments.git  
cd GamesENGII
```

2. Configure Git identity (one-time)

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

3. Create your personal folder (first lab)

As mentioned in the Git repository, the Weekly lab must be submitted under the respective folder.

For instance, the root folder will be **LABNN** (where, NN, will be replaced with the Lab numbers, such as Lab01, Lab02, Lab03)

4. Add GIT Collaborator

Please add me (**oa-SETU**) / (omer.ali@setu.ie) as a collaborator to your repositories.

This will allow me to look at your submissions and mark your labs.

Final Notes

- Keep commits clean and meaningful.
- You are expected to commit after every task/objective has been completed (*Remember, committing regularly and with incremental steps is good practice and helps to evaluate, rollback and reconcile in a production environment*).
- Your folder = your responsibility.

Happy coding 🎮

Contact

Availability: **C202** (Wednesday 1500 - 1700 hrs)

For more info/help , email me at omer.ali@setu.ie